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Data assetization, financial distress and diversification: A moderated mediation model

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ABSTRACT

Amidst intensifying competition, Chinese firms are thriving by leveraging data assets and corporate diversification to propel innovation and solidify their competitive edge. This study constructs a moderated mediation model to examine the effect of data assetization on diversification. Empirical results indicate that data assetization significantly promotes the implementation of diversified strategies, with the reduction of financial distress risk playing a mediating role. Dynamic capabilities demonstrate dual moderating effects: innovation capability and absorptive capacity strengthen the impact, while adaptive capability negatively moderates the relationship. Heterogeneity analysis reveals that data assetization more strongly drives diversification in technology-intensive firms and those in competitive industries. Tests on economic consequences confirm that data assetization-driven diversification effectively enhances organizational resilience.

1. Introduction

Data has become a key driver for enhancing the effectiveness of the digital economy, digital industrialization, and digital governance (Zhang and Du, 2025). As a vital pathway for promoting data circulation and value creation, data assetization transforms data resources into quantifiable, tradable strategic assets through processes such as rights confirmation, governance, and circulation (Liu et al., 2025), thereby generating future economic benefits and enhancing decision-making (Bonvino and Giorgino, 2024). Data assetization has become a critical engine for cultivating new business strategy. Firms can optimize operational efficiency, innovate business models, and create incremental value in fields such as financial risk control and medical diagnosis (Yu et al., 2025; Zhou et al., 2026).

Regarding theoretical research, first, data assetization optimizes capital market performance. It not only reduces information asymmetry but also serves as an opportunity to gain favorable attention from stakeholders (Ma et al., 2023). This effectively broadens financing channels and alleviates financing constraints. Second, it enhances operational efficiency. Data-driven insights help explore diversified markets and build differentiated competitiveness, ultimately strengthening corporate resilience and unlocking development potential (Ma et al., 2024). Finally, it enables green and sustainable development. Leveraging the reusability and shareability of data resources, it serves as a green transition path that breaks traditional resource constraints, representing a key practice in social and environmental responsibility (Yang et al., 2023).

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Although data assetization has garnered increasing scholarly attention, its influence on business strategy remains largely unexamined. On one hand, as a value-appreciating resource under corporate control (Ji et al., 2025), data assets provide the economic foundation for business expansion. On the other hand, data assetization enables dynamic capability building through deeper integration of production factors (Tang et al., 2025). Therefore, considering the dual function of data assetization in capital reinforcement and capability building, this paper investigates its impact on business diversification and further explores the underlying mechanisms.

Relative to prior research, the potential marginal contributions include: First, the study identifies data assetization as an economic driver for diversification, thereby advancing the understanding of data's transition from concept to strategic asset. Second, it elucidates the specific mechanisms by which data assetization influences strategy, revealing financial distress as a mediator and organizational resilience as an outcome of diversification. Finally, it reveals the dual moderating role of dynamic capabilities, expanding the understanding of boundary conditions for realizing data value.

2. Theoretical analysis and research hypotheses

2.1. Data assetization and diversification

Grounded in production factor allocation theory, data factors possess a multiplier effect and are reusable (Wang et al., 2021). They integrate with traditional factors to catalyze innovation, enabling firms to upgrade offerings and consolidate competitive advantages (Sun and Du, 2024), thereby providing financial support for diversification. Data assetization reflects a shift toward digital-intelligent business transformation (Zhao et al., 2024). It enables accurate consumer profiling, reduces marketing costs, and employs intelligent models to predict the outcomes of diversification. From a green governance standpoint, the green and innovative attributes of data assets enhance social responsibility profiles (Liu et al., 2026), thereby supplying innovative and green pathways for diversification. Data assets heighten sensitivity to external factors like policies and technologies, enhancing resilience to market volatility, and strengthens supply chain robustness (Zamani et al., 2023).

The following hypothesis is proposed:

H1. Data assetization significantly promotes diversification.

2.2. The mediating role of financial distress

Data assetization helps broaden financing channels and lower financing costs. Rights-confirmed and value-assessed data assets function as new collateral, boosting credit access and creating innovative financing options (Wang, 2026). They help curb speculative investments and lower capital allocation risks, driving optimization of cost structures and improvement of cash flows. By expanding marketing channels and reducing costs, data enables early prediction and timely control of operational risks (Einbinder et al., 2025). As recognized balance sheet items, data assets enhance total assets, reduce leverage, and optimize financial statement structure, sending positive signals to stakeholders through improved metrics (Qian et al., 2025).

Based on the financial flexibility theory, financial robustness provides firms with sufficient capital support. The initial phase of diversification involves substantial sunk costs (Zhou, 2011). Firms should possess adequate cash flow and strong financing capacity for nurturing new businesses. Meanwhile, a stable financial environment enhances a firm's resilience to risk (Kyrdoda et al., 2023). Financially robust firms are better positioned to endure experimentation and fund the necessary waiting period for market alignment (Shahbaz et al., 2021). Sound financial conditions exhibit greater flexibility and efficiency in resource allocation (Chen et al., 2025), so they create synergistic effects across different business lines and enable timely responses to market opportunities.

The following hypothesis is proposed:

H2. Data assetization promotes corporate diversification by alleviating financial distress and enhancing financial robustness.

2.3. The moderating role of dynamic capabilities

Dynamic capabilities comprise three dimensions: innovation capability, absorptive capacity, and adaptive capability (Wang and Ahmed, 2007). They serve as a source of competitive advantage for firms, and enable firms to develop data-driven businesses based on native data processes (Augier and Teece, 2009), thereby shaping competitive positioning (Ciampi et al., 2021).

2.3.1. Innovation capability

Based on Schumpeter innovation theory, innovation capability is conceptualized as the synthesized competitive capability forged through novel combinations of production factors such as knowledge, intellect, and skills. Firms with strong innovation capability can more effectively extract value from data (Chatterjee et al., 2024). By analyzing consumer demand, developing new product lines, or creating cross-sector services, innovation capability enhances data efficiency and explores high-value-added businesses. Innovation capability contributes to firms' resilience to risks (Kyrdoda et al., 2023). Data assetization optimizes resource allocation and reduces operating costs, thereby easing financial distress. Stronger innovation capability empowers firms to reduce costs through intelligent supply chains and strategic alliances (Zhang et al., 2024), lowering the likelihood of financial distress or facilitating a quicker recovery from it.

The following hypothesis is proposed:

H3a. Innovation capability positively moderates the effects of data assetization on both diversification and financial distress alleviation.

2.3.2. Absorptive capacity

According to absorptive capacity theory, stronger absorptive capacity facilitates more effective integration of internal data and external information. It can rapidly interpret and apply relevant data, and leverage data intelligence methods to generate actionable insights, facilitate deep integration between online and offline operations (Lee et al., 2024), contributing to diversification. Additionally, firms with high absorptive capacity can quickly absorb industry best practices, avoid inefficient investments (Mu and Jiang, 2024), accurately analyze market demand and cost structures (Fuad et al., 2024). This optimizes the allocation of supply chain resources, thereby alleviating cash flow pressure (McEvily et al., 2004).

The following hypothesis is proposed:

H3b. Absorptive capacity positively moderates the effects of data assetization on both diversification and financial distress alleviation.

2.3.3. Adaptive capability

Consistent with organizational behavior theory, firms possessing strong digital capabilities and financial stability tend to pursue diversification (Chen et al., 2019). However, excessive adaptive capability often leads to strategic shifts (Cavusgil and Deligonul, 2025). This instability not only scatters efforts and dilutes core competitiveness (Zahra et al., 2022). Therefore, it impedes the effective translation of data assets into diversified businesses and undermines the ability to leverage financial health for sustained diversification support. In effect, stronger adaptive capability weakens the positive impact of data assetization and financial health on diversification.

The following hypothesis is proposed:

H3c. Adaptive capability negatively moderates the effects of data assetization and financial distress on diversification.

2.4. Economic consequences of data assetization on diversification

From the resource-based view, data assets can lower cognitive barriers and information asymmetry when entering new markets (Qian et al., 2025). Data-driven refined management and cross-unit coordinated analysis enhance the identification of inter-business risk linkages, and diversification enabled by data empowerment essentially establishes a more resilient resource portfolio buffer (Helfat et al., 2023). Data assets enhance the ability to perceive and respond to environmental changes, enabling firms to cope with environmental uncertainty and risk, reallocate resources agilely, and recover rapidly from disruptions (Goldfarb and Tucker, 2019; Chen et al., 2025), thereby leading to greater impact resistance, faster recovery, and improved adaptability.

The following hypothesis is proposed:

H4. Enhancing organizational resilience is an economic consequence of data assetization through diversified operations.

The theoretical framework is presented in Fig. 1.

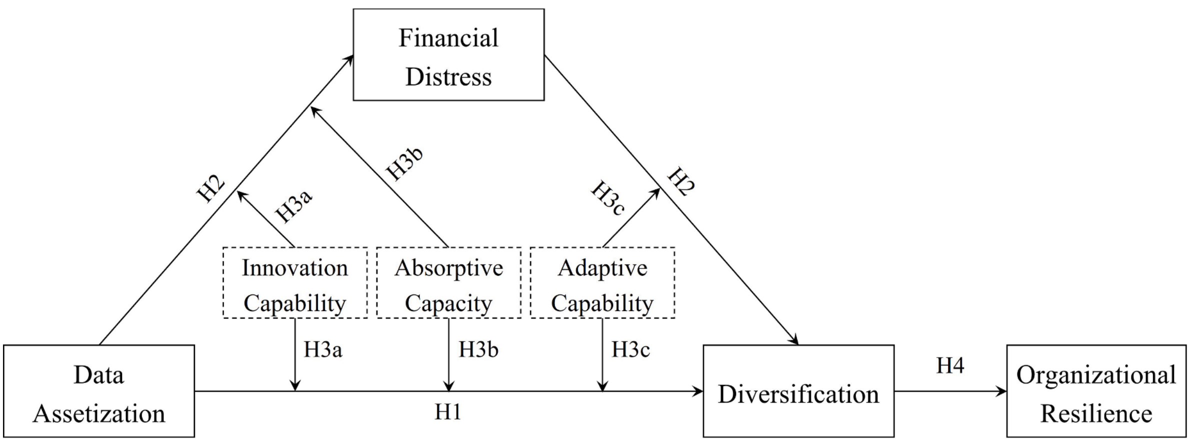

Fig. 1. Theoretical framework.

3. Research design

3.1. Sample selection and data sources

China's National Big Data Strategy started in 2015, and significantly accelerated digital transformation and data assetization. Shanghai and Shenzhen A-share listed companies from 2015 to 2023 are selected as the initial sample. The following procedures are applied:

- Exclusion of firms in the financial and insurance industries;
- Removal of samples labeled as ST, *ST or PT during the research period;
- Elimination of observations with significantly missing data;
- All continuous variables were winsorized at 1 % level.

The final sample consists of 31,599 observations. Data are primarily sourced from the CSMAR and WIND databases.

3.2. Variable measurement

3.2.1. Independent variable: data assetization (DA)

The study complies a corpus of legal and regulatory texts on data assets, employing "data" "digital" "network" and "information" as seed words. The Word2Vec neural network model is employed to construct similar word sets. Python is used to analyze annual reports and calculate word frequencies (Liu et al., 2025). Self-use data assets (*SDA*) and transactional data assets (*TDA*) are used for detailed analysis.

3.2.2. Dependent variable: diversification (DS)

This study calculates the revenue entropy index as the dependent variable (Fuente et al., 2025). Herfindahl-Hirschman Index (*HHI*) is employed in robustness tests. The revenue entropy index is a positive indicator, and *HHI* is a negative indicator.

3.2.3. Mediating variable: financial distress (FD)

This study uses the Z-Score as a proxy indicator for financial distress (Yin et al., 2023), and a higher value indicates a lower probability of falling into financial distress.

3.2.4. Moderating variables: dynamic capabilities (DC)

Dynamic capabilities comprise innovation capability (*INN*), absorptive capacity (*ABS*) and adaptive capability (*ADA*) (Li et al., 2024). To align adaptive capability with innovation capability and absorptive capacity as positive indicators, adaptive capability is assigned a negative value.

3.2.5. Control variables

Consistent with research topic and prior studies, firm size (*Size*), asset-liability ratio (*Lev*), return on equity (*ROE*), state ownership

Table 1
Definition and Explanation of Main Variables.

Type	Name	Symbol	Description
Independent Variable	Data assetization	*DA*	ln (Frequency of data assetization+1)
	Self-use data assets	*SDA*	ln (Frequency of self-use data assetization+1)
	Transactional data assets	*TDA*	ln (Frequency of transactional data assetization+1)
Dependent Variable	Diversification strategy	*DS*	Revenue Entropy Index
Mediating Variable	Financial distress	*FD*	Z-score
Moderating Variable	Dynamic capabilities	*DC*	
	Innovation capability	*INN*	R&D Intensity and Technical Staff Ratio
	Absorptive capacity	*ABS*	R&D-to-Revenue Ratio
	Adaptive capability	*ADA*	The negative value of coefficient of variation of R&D, capital, and advertising expenditure
Control Variable	Firm size	*Size*	Natural logarithm of total assets
	Debt to asset ratio	*Lev*	Total liabilities / Total assets
	Return on Equity	*ROE*	Net profit / Average balance of owners' equity
	Ownership type	*SOE*	State-owned firm dummy
	CEO duality	*Dual*	CEO duality dummy
	List age	*List Age*	ln (Current year-Listing year+1)
Other Variables	Instrumental variables for Data assetization in 4.3.1	*LDA*	Industry average value of data assetization
	Instrumental variables for Data assetization in 4.3.2	*RDL*	ln (regional software revenue+1)
	Organizational Resilience	*OR*	Entropy-based composite score (stock return volatility & 3-year sales growth)

(*SOE*), CEO-chair duality (*Dual*), and listing age (*List Age*) are selected as control variables.

3.2.6. Other variables

To address endogeneity, the leave-one-out industry average (*LDA*) and regional digital intelligence level (*RDL*) are selected as instrumental variables. Organizational resilience (*OR*) is measured following the approach of Wu et al. (2025).

Detailed definitions and descriptions of all variables are provided in Table 1.

3.3. Model specification

3.3.1. Main effect model

To examine the impact of data assetization on corporate diversification, a regression model is constructed with year and industry fixed.

$$DS_{i,t} = \alpha_0 + \alpha_1 DA_{i,t} + \sum Controls_{i,t} + \sum Year + \sum Ind + \varepsilon_{i,t} \tag{1}$$

In Model (1), $Controls_{i,t}$ refers to the set of control variables, and $\varepsilon_{i,t}$ is the error term. All regressions, including those in subsequent models, employ robust standard errors clustered at the firm level.

3.3.2. Moderated mediation model

To test the moderating effect of dynamic capabilities on the relationship between data assetization and diversification:

$$DS_{i,t} = \beta_0 + \beta_1 DA_{i,t} + \beta_2 DC_{i,t} \times DA_{i,t} + \beta_3 DC_{i,t} + \sum Controls_{i,t} + \sum Year + \sum Ind + \varepsilon_{i,t} \tag{2}$$

In Model (2), a significant β_2 indicates that dynamic capabilities play a moderating role.

To examine the moderating role of dynamic capabilities on the relationship between data assetization and financial distress:

$$FD_{i,t} = \gamma_0 + \gamma_1 DA_{i,t} + \gamma_2 DC_{i,t} \times DA_{i,t} + \gamma_3 DC_{i,t} + \sum Controls_{i,t} + \sum Year + \sum Ind + \varepsilon_{i,t} \tag{3}$$

In Model (3), if the mediation effect holds and γ_1 is significant, a significant γ_2 would suggest that moderating role of dynamic capabilities.

To test the moderating effect of dynamic capabilities on the relationship between financial distress and diversification:

If the direct effect in Model (2) is not moderated:

$$DS_{i,t} = \theta_0 + \theta_1 DA_{i,t} + \theta_2 DC_{i,t} + \theta_3 FD_{i,t} + \theta_4 DC_{i,t} \times FD_{i,t} + \sum Controls_{i,t} + \sum Year + \sum Ind + \varepsilon_{i,t} \tag{4}$$

In Model (4), a significant θ_4 indicates a moderating role of dynamic capabilities.

If the direct effect in Model (2) is moderated:

$$DS_{i,t} = \eta_0 + \eta_1 DA_{i,t} + \eta_2 DC_{i,t} + \eta_3 DC_{i,t} \times DA_{i,t} + \eta_4 FD_{i,t} + \eta_5 DC_{i,t} \times FD_{i,t} + \sum Controls_{i,t} + \sum Year + \sum Ind + \varepsilon_{i,t} \tag{5}$$

In Model (5), the mediation effect is supported by significant η_4, and a significant η_5 would imply that dynamic capabilities moderate the effect of financial distress on diversification.

4. Empirical analysis

4.1. Benchmark regression results

Table 2 shows the benchmark regression results without and with control variables respectively, the coefficients of data assetization remain statistically significant at the 1 % level, indicating that higher degree of data assetization significantly facilitates diversification. Hypothesis H1 is supported.

Table 2
Benchmark Regression Results.

Variables	(1) DS	(2) DS
DA	0.113***	0.121***
	(0.008)	(0.007)
Constant	−0.003	−0.029***
	(0.005)	(0.007)
Industry	YES	YES
Year	YES	YES
Controls	-	YES
N	31,599	31,599
Adj R^2	0.0927	0.1513

*** represent statistical significance at the 1 % levels.

4.2. Robustness tests

4.2.1. Alternative variable measurements

As shown in column (1) of Table 3, diversification is measured by the Herfindahl-Hirschman Index, where a higher *HHI* indicates lower diversification. In column (2)-(3), data assetization is categorized into self-use data assets (*SDA*) and transactional data assets (*TDA*). To address potential reverse causality and account for lagged effects, column (4) shows results using the one-period lagged data assetization variable (L.*DA*). The results confirm the robustness of H1.

4.2.2. Changing fixed effects

Column (1)-(3) of Table 4 present regression results based on broad industry fixed effects, firm fixed effects, and industry-year interactive fixed effects respectively, further validating the robustness of H1.

4.3. Endogeneity tests

4.3.1. Instrumental variable approach

To address potential endogeneity from omitted variables, a two-stage least squares (*2SLS*) approach is employed. As shown in Table 5, the instrumental variable is strongly correlated with data assetization in the first stage. The second-stage results confirm a statistically significant positive effect of data assetization on corporate diversification after mitigating endogeneity, which supports Hypothesis H1.

4.3.2. Heckman two-stage method

To address potential endogeneity from sample selection bias, the Heckman two-stage procedure is applied, constructing a binary (0–1) indicator variable (*DAR*) based on *DA*, and using the regional digitalization level (*RDL*) as the instrument. The results in Table 6 show that the instrument is significant in the first stage, while the insignificant inverse Mills ratio in the second stage suggests no severe sample selection bias. The significant positive effect of data assetization on diversification further supports Hypothesis H1.

5. Further analysis

5.1. Heterogeneity analysis

Columns (1)-(3) in Table 7 report results for technology-, capital-, and labor-intensive firms, and columns (4)-(5) for firms in competitive and regulated industries. Regulated industries refer to sectors either state-dominated or subject to stringent government regulation, such as healthcare, education, and finance.

Among factor intensity categories, the promoting effect is stronger in technology-intensive firms than in labor-intensive ones. For technology-intensive firms, they align more closely with data-driven and intelligent iterative processes, thereby effectively leveraging data assetization for product innovation and market expansion.

The effect of data assetization is stronger in competitive industries. These industries can respond swiftly to market changes, utilizing data resources to optimize product portfolios and drive diversification. In contrast, regulated industries face stronger policy constraints, data assetization in these sectors often focuses more on compliance rather than strategic innovation.

Table 3
Results after Substituting Variables.

Variables	(1) DS-HHI	(2) DS	(3) DS	(4) DS
DA	−0.122***			
	(0.007)			
SDA		0.114***		
		(0.007)		
TDA			0.103***	
			(0.007)	
L.DA				0.119***
				(0.008)
Constant	0.028***	−0.029***	−0.026***	−0.025***
	(0.008)	(0.007)	(0.007)	(0.008)
Industry	YES	YES	YES	YES
Year	YES	YES	YES	YES
Controls	YES	YES	YES	YES
N	31,599	31,599	31,599	26,397
Adj R^2	0.1437	0.1507	0.1504	0.1467

*** represent statistical significance at the 1 % levels.

Table 4
Results after Changing Fixed Effects.

Variables	(1) DS	(2) DS	(3) DS
DA	0.114***	0.044***	0.120***
	(0.007)	(0.007)	(0.007)
Constant	−0.032***	−0.009	−0.029***
	(0.008)	(0.038)	(0.008)
Year	YES	YES	YES
Broad Industry	YES	-	-
Id	-	YES	-
Industry	-	-	YES
Industry*Year	-	-	YES
Controls	YES	YES	YES
N	31,599	31,354	31,599
Adj R^2	0.1124	0.8171	0.1412

*** represent statistical significance at the 1 % levels.

Table 5
Results of Instrumental Variable Method.

Variables	(1) DA	(2) DS
DA		0.143**
		(0.062)
LDA	0.882***	
	(0.052)	
Industry	YES	YES
Year	YES	YES
Controls	YES	YES
N	31,568	31,568
Kleibergen-Paap Wald rk F	283.03 > 16.38 (Stock-Yogo weak ID test, 10 % maximal IV size)	
Kleibergen-Paap rk LM	263.326, $p = 0.000$	
R^2		0.0712

*** and ** represent statistical significance at the 1 %, 5 % levels, respectively.

Table 6
Results of Heckman's Two Stages.

Variables	(1) DAR	(2) DS
DA		0.177***
		(0.018)
IMR		−0.075
		(0.180)
RDL	0.041***	
	(0.011)	
Constant	−1.235***	−0.053
	(0.134)	(0.017)
Industry	YES	YES
Year	YES	YES
Controls	YES	YES
N	31,419	7076
R^2	0.2967	0.1455

*** represent statistical significance at the 1 % levels.

5.2. *Mediating effect*

Column (2) in Table 8 reports the effect of data assetization on financial distress, the coefficient of data assetization is significantly positive, implying that data assetization reduces the likelihood of firms experiencing financial distress.

Column (3) confirms the mediating role of financial distress, indicating that data assetization lowers financial distress and enhances financial robustness, ultimately facilitates diversification. Hypothesis H2 is supported.

Table 7
Heterogeneity Grouping Test Results.

Variables	DS-Factor Intensity Categories			DS-Competitive/Regulatory Industries	
	(1) DS-TEC	(2) DS-CAP	(3) DS-LAB	(4) DS-COM	(5) DS-REG
DA	0.162***	0.111***	0.063***	0.131***	0.070***
	(0.010)	(0.021)	(0.013)	(0.008)	(0.016)
Constant	−0.092***	−0.057***	0.055***	−0.021**	0.001
	(0.011)	(0.020)	(0.013)	(0.009)	(0.017)
Industry	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES
N	15,701	5339	10,248	24,356	7243
Adj R^2	0.1073	0.1296	0.2014	0.1308	0.2180
P-value for group difference	0.000			0.000	

*** and ** represent statistical significance at the 1 %, 5 % levels, respectively; P-values for coefficient differences are derived from Fisher's combination test bootstrapped 1000 times.

Table 8
Results of the Mediating Effect.

Variables	(1) DS	(2) FD	(3) DS
DA	0.121***	0.023***	0.120***
	(0.007)	(0.006)	(0.007)
FD			0.017**
			(0.007)
Constant	−0.029***	−0.015**	−0.029***
	(0.007)	(0.006)	(0.007)
Industry	YES	YES	YES
Year	YES	YES	YES
Controls	YES	YES	YES
N	31,599	31,599	31,599
Adj R^2	0.1513	0.4462	0.1514

*** and ** represent statistical significance at the 1 %, 5 % levels, respectively.

5.3. Moderated mediation effect

Columns (1)-(3) in Table 9 report results for the moderating role of innovation capability. The significant interaction in column (1) indicates its positive moderating effect on the direct relationship. Column (2) shows that innovation capability also strengthens the effect of data assetization on financial distress. Column (3) suggests no moderating effect in the second stage of the mediation. Therefore, H3a is supported.

Similarly, columns (4)-(6) present results for absorptive capacity, demonstrating a similar moderating pattern to innovation capability, supporting H3b.

Columns (7)-(9) present the results for adaptive capability. The significant interaction in column (7) shows its negative moderating effect on the direct relationship. Column (8) indicates no significant moderation in the first stage of mediation, and column (9) confirms a significantly negative interaction with financial distress, supporting its negative moderating role in the second stage. Therefore, H3c is supported.

5.4. Economic consequences

As shown in Table 10, data assetization has a direct positive effect on *OR* and strengthens it through diversification, supporting H4.

5.5. Economic consequences under internal and external heterogeneity

5.5.1. Factor intensity types

As presented in Table 11, data assetization enhances organizational resilience through promoting diversification across all subsamples. Column (1)-(3) indicates full mediation by diversification, suggesting that data-driven diversification is a primary source of organizational resilience enhancement in technology-intensive firms.

5.5.2. Competitive vs. regulated industries

As shown in Table 12, in competitive industries, data assetization significantly enhances organizational resilience by promoting diversification. However, this effect is not observed in regulated industries, indicating that in regulated industries, the enhancement of

Table 9
Results of the Moderated Mediation Effect.

Variables	DS-Innovation Capability			DS-Absorptive Capacity			DS-Adaptive Capability		
	(1) DS	(2) FD	(3) DS	(4) DS	(5) FD	(6) DS	(7) DS	(8) FD	(9) DS
DA	0.115***	0.020***	0.115***	0.125***	0.021***	0.125***	0.121***	0.019***	0.120***
	(0.008)	(0.006)	(0.008)	(0.007)	(0.006)	(0.007)	(0.007)	(0.006)	(0.007)
DC	−0.050***	−0.010	−0.051***	−0.093***	−0.003	−0.093***	0.007	0.050***	0.006
	(0.008)	(0.007)	(0.008)	(0.008)	(0.007)	(0.008)	(0.006)	(0.005)	(0.006)
DA*DC	0.041***	0.013***	0.041***	0.033***	0.008**	0.033***	−0.011*	−0.000	−0.010
	(0.005)	(0.004)	(0.005)	(0.005)	(0.004)	(0.005)	(0.006)	(0.005)	(0.006)
FD			0.016**			0.017**			0.017**
			(0.007)			(0.007)			(0.007)
DC*FD			0.008			0.000			−0.019***
			(0.005)			(0.005)			(0.005)
Constant	−0.049***	−0.021***	−0.049***	−0.042***	−0.018***	−0.041***	−0.028***	−0.015**	−0.024***
	(0.008)	(0.006)	(0.008)	(0.008)	(0.006)	(0.008)	(0.008)	(0.006)	(0.008)
Industry	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	31,599	31,599	31,599	31,599	31,599	31,599	31,599	31,599	31,599
Adj R^2	0.1533	0.4463	0.1535	0.1548	0.4462	0.1549	0.1513	0.4479	0.1518

*** and ** represent statistical significance at the 1 %, 5 % levels, respectively.

Table 10
Economic Consequences: Organizational Resilience.

Variables	(1) OR	(2) DS	(3) OR
DA	0.008***	0.121***	0.007***
	(0.002)	(0.007)	(0.002)
DS			0.011***
			(0.001)
Constant	0.503***	−0.029***	0.503***
	(0.002)	(0.007)	(0.002)
Industry	YES	YES	YES
Year	YES	YES	YES
Controls	YES	YES	YES
N	31,599	31,599	31,599
Adj R^2	0.5541	0.1513	0.5550

*** represent statistical significance at the 1 % levels.

Table 11
Economic Consequences Across Factor Intensive Types.

Variables	OR-Technology Intensive			OR-Capital Intensive			OR-Labor Intensive		
	(1) OR	(2) DS	(3) OR	(4) OR	(5) DS	(6) OR	(7) OR	(8) DS	(9) OR
DA	0.005*	0.162***	0.003	0.011**	0.111***	0.010**	0.012***	0.063***	0.012***
	(0.003)	(0.010)	(0.003)	(0.005)	(0.021)	(0.005)	(0.003)	(0.013)	(0.003)
DS			0.014***			0.010***			0.007***
			(0.002)			(0.003)			(0.002)
Constant	0.517***	−0.092***	0.519***	0.506***	−0.057***	0.507***	0.500***	0.055***	0.500***
	(0.003)	(0.011)	(0.003)	(0.004)	(0.020)	(0.004)	(0.003)	(0.013)	(0.003)
Industry	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	15,701	15,701	15,701	5339	5339	5339	10,248	10,248	10,248
Adj R^2	0.5019	0.1073	0.5032	0.5993	0.1296	0.6001	0.6151	0.2014	0.6155

***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 12
Economic Consequences Across Competitive/Regulatory Industries.

Variables	OR-Competitive Industries			OR-Regulatory Industries		
	(1) OR	(2) DS	(3) OR	(4) OR	(5) DS	(6) OR
DA	0.010***	0.131***	0.008***	0.001	0.070***	0.000
	(0.002)	(0.008)	(0.002)	(0.003)	(0.016)	(0.003)
DS			0.009***			0.012***
			(0.002)			(0.002)
Constant	0.513***	−0.021**	0.513***	0.496***	0.001	0.496***
	(0.002)	(0.009)	(0.002)	(0.004)	(0.017)	(0.004)
Industry	YES	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
N	24,356	24,356	24,356	7243	7243	7243
Adj R^2	0.5314	0.1308	0.5319	0.6503	0.2180	0.6514

*** and ** represent statistical significance at the 1 %, 5 % levels, respectively.

organizational resilience does not depend on specific data assetization pathways.

6. Conclusion

The study empirically demonstrates that data assetization among Chinese A-share listed firms promotes business diversification, with financial distress serving as a mediator and dynamic capabilities acting as a dual moderator. Furthermore, data assetization is found to enhance organizational resilience through its positive impact on diversification. These effects vary across industries with different factor intensity types and regulatory environments.

Relative to extant research, theoretically, this study establishes data assetization as a strategic driver of diversification, extending its conceptualization beyond a technical asset. By unpacking the mechanisms linking data assetization to strategic outcomes, it bridges the data value literature and research on corporate strategy and resilience. Examining the roles of dynamic capabilities and financial distress, this study identifies boundary conditions and underlying mechanisms, providing insights into data value realization. From a practical standpoint, managers should recognize that the strategic value of data as an asset is amplified when its utilization is coupled with dynamic capability development and tailored to specific industry contexts. For policymakers, the design of data-related regulations should reflect distinct industry characteristics, with the aim of effectively enabling data-driven strategic transformation.

Regarding limitations, the measurement of data assetization relies on annual reports, lacking qualitative inputs such as expert evaluations. Furthermore, while financial distress risk and dynamic capabilities are identified as key mechanisms, other important factors—such as data governance maturity and supply chain position—remain unexplored. Future research could address these gaps by developing more refined measures of data assetization that incorporate qualitative inputs, extending the sample to private firms, exploring alternative mechanisms, and employing longitudinal designs to better capture causal dynamics.

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CRediT authorship contribution statement

Yu Zhang: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jinsong Zhang:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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